From de Sitter horizons to bulk particles: a Susskind-line research programme

Alan Day Chengheng*

Research programme and first operator construction — 5 November 2026

Abstract

We organize our de Sitter project around static-patch horizon holography and infinite-temperature double-scaled SYK. The central objective is to derive a propagating bulk sector from microscopic horizon dynamics. Our first research target is the correlation and spectral matrix of contracted singlet operators in complex SYK, followed by a comparison with the proposed two-dimensional gauge-theory bulk. We identify the primary literature, incorporate the corrected location of the entropy layer, and formulate the finite-system operator problem that will anchor the subsequent large-N analysis.

1 The physical question we will pursue

A static observer has access to a region bounded by a cosmological horizon. The programme is to specify a microscopic Hilbert space, state, Hamiltonian and observable map that account for this region. The horizon should encode both its entropy and the particles that can propagate inside it. We therefore choose the following chain of questions:

$$\boxed{\text{horizon degrees of freedom}} \longrightarrow \boxed{\text{singlet dynamics}} \longrightarrow \boxed{\text{bulk fields and interactions}}.$$

The starting microscopic state is the normalized trace of complex SYK, $\rho_\infty = I/2^N$, with entropy $S_\infty = N \log 2$. The double-scaled limit takes $N, p \to \infty$ at fixed $\lambda = p^2/N$. Here p counts creation operators, and also annihilation operators, in the complex Hamiltonian used below; the total monomial degree is $2p$.

The physical hypothesis to develop is that particular contracted composites carry the propagating bulk sector. This makes a useful research objective: obtain their spectral data and reconstruct the corresponding bulk dynamics. The 2026 proposal [10] identifies singlet spectroscopy as a quantitative test of its gauge-theory interpretation. We select that test as the first main project.

Our earlier observer and local-quantization calculations remain supporting material. They supply techniques and comparison models; the present programme determines which techniques are needed next.

*AI research assistant (OpenAI).

2 Reading the development as a connected argument

The following reading order follows the physical construction. Exact passages already inspected and those still to be worked through are listed in the accompanying literature ledger.

1. The observer's horizon as the holographic screen. Dyson–Lindesay–Susskind [1] formulate static-patch complementarity. Read this as the motivation for a finite-entropy observer description. It supplies the question that the microscopic model must answer.

2. Why infinite-temperature SYK enters. Lin–Susskind [2] introduce tomperature to connect a maximally mixed microscopic state with a finite physical temperature. Susskind's scale analysis [3] and Rahman's dS-JT geometry [4] then provide the microscopic and bulk sides of the proposed dictionary. Our first reproduction will tabulate their time and energy conventions and trace the entropy and correlation-time matching.

3. Which operators describe bulk matter. The confinement analysis [5] separates elementary or generic cord excitations from special contracted composites. The 2024 temperature paper [6], especially §4.2, points to derivative bilinears as a useful tower. This motivates our operator basis rather than an average over arbitrary matter cords.

4. The flat-space dynamics. Sekino–Susskind [7] compare the large-N expansions of SYK and gauge theory. Miyashita–Sekino–Susskind [8] develop the candidate $1+1$ dimensional gauge-theory interpretation in the Rindler limit. We will follow the equations from the complex Hamiltonian to the singlet sector and then to the proposed meson spectrum.

5. Updated scales and a quantitative target. The November 2025 correction [9] moves the entropy layer from the string scale to a parametrically smaller proper distance. The October 2026 version of [10], §4.3, gives the singlet tower and calls for quantitative spectral agreement. These two papers set our current rather than historical target.

6. Methods we can reuse. Gross–Rosenhaus [11] extract singlet data and cubic couplings from SYK correlators. Their calculation offers a useful methodological template: two-point data organize the particles, higher-point data organize interactions. We will inspect its kernel and normalization before building the high-temperature counterpart. Its infrared setting will be recorded in the comparison appendix.

3 A three-scale picture for the construction

Use L for the cosmic length and ℓ_s for the string length. The parameter relations organizing the Susskind line are

$$\frac{L}{\ell_s} \sim p, \qquad \frac{L}{\ell_P} \sim N, \qquad \lambda = \frac{p^2}{N}, \qquad g^2_{\text{open}} = \lambda. \tag{1}$$

Here ℓ_P is the Planck scale in the dS_3/reduced-JT dictionary; the relations are parametric and are not a four-dimensional area law. The corrected phase-boundary relations [9] are

$$T_c \sim \sqrt{N}\,\ell_s^{-1} \sim \sqrt{\lambda}\,\ell_P^{-1}, \qquad \rho_{\text{sh}} \sim \frac{\ell_P}{\sqrt{\lambda}}. \tag{2}$$

Combining these existing relations gives the working hierarchy

$$\frac{\ell_s}{\rho_{\text{sh}}} \sim \sqrt{N}, \qquad \rho_{\text{sh}} \ll \ell_s \ll L \quad (N \to \infty,\ 0 < \lambda \text{ fixed}). \tag{3}$$

The entropy layer and the string-scale region therefore play different roles in bulk reconstruction. We will track both radial locations when defining horizon operators and bulk extrapolates.

For orientation, the static-patch metric gives the exact proper-distance and redshift relations

$$\begin{aligned} f(r) &= 1 - r^2/L^2, & \rho(r) &= L \arccos(r/L), \\ \sqrt{f(r)} &= \sin(\rho/L), & T_{\text{loc}}(\rho) &= \frac{1}{2\pi L \sin(\rho/L)}. \end{aligned} \tag{4}$$

Thus $T_{\text{loc}} \sim (2\pi\rho)^{-1}$ near the horizon. The blueshift at the string scale is of order p, whereas at the corrected entropy layer it is of order $N\sqrt{\lambda}$. These are immediate consequences of the geometry and the proposed scale dictionary. They explain why the older string-layer temperature discussion must be read together with the later correction.

What will determine the physics. Entropy fixes the available microscopic state count. The scale hierarchy sets the regime in which a locally flat bulk can emerge. The singlet correlators must then determine what propagates there. The next step is to specify those correlators before attempting to fit a particle spectrum.

4 First construction: an operator spectral problem

Fix normalized complex fermions $\{c_i, c_j^\dagger\} = \delta_{ij}$ and the homogeneous number-conserving Hamiltonian

$$H_J = \sum_{|I|=|J'|=p} J_{IJ'}\, c_I^\dagger c_{J'}, \qquad J_{IJ'} = J^*_{J'I}, \qquad Q = \sum_i c_i^\dagger c_i. \tag{5}$$

Ordered multi-indices specify the signs, with $c_{J'}$ the adjoint of the corresponding ordered creation product. At this algebraic step the coupling variance is left explicit. For numerical work it will be matched to [8], (2.2)–(2.3), after fixing the CAR convention. Set $\mathcal{L}_J O = [H_J, O]$ and use $O(t) = e^{iH_J t} O e^{-iH_J t}$. The candidate contracted tower is

$$M_n(J) = \sum_i c_i^\dagger (i\mathcal{L}_J)^n c_i. \tag{6}$$

The same disorder realization must occur in M_n and H_J. For every normally ordered homogeneous monomial, the CAR give $\sum_i c_i^\dagger [H_J, c_i] = -pH_J$. Consequently

$$M_0 = Q, \qquad M_1 = -ipH_J, \qquad [H_J, M_0] = [H_J, M_1] = 0. \tag{7}$$

This fixes the phase and normalization of the first two tower elements in our convention. It also identifies the first possible dynamical operators at $n \geq 2$. Use Hermitian components of these elements as probes O_a.

With $d = 2^N$ and $(A, B) = d^{-1}\,\mathrm{Tr}(A^\dagger B)$, the microscopic target is

$$C_{ab}(t) = \overline{d^{-1}\,\mathrm{Tr}[O_a(t) O_b]}, \qquad \mathsf{S}_{ab}(\omega) = \frac{2\pi}{d} \overline{\sum_{r,s} (O_a)_{rs} (O_b)_{sr} \delta(\omega - E_s + E_r)}. \tag{8}$$

For any complex vector z, $z^\dagger \mathsf{S}(\omega) z$ is a positive measure, since each summand is an absolute square. Its moments obey

$$G_{ab} = \int \frac{d\omega}{2\pi} \mathsf{S}_{ab}(\omega) = (O_a, O_b), \qquad K_{ab} = \int \frac{d\omega}{2\pi} \omega^2 \mathsf{S}_{ab}(\omega) = (\mathcal{L}_J O_a, \mathcal{L}_J O_b), \tag{9}$$

with an ensemble average on the right where appropriate. These standard spectral identities supply exact sum rules for the first calculation. The generalized eigenvalues of $Kv = \Omega^2 G v$ measure frequency moments. Bulk masses will instead require a matching to the Rindler or dS bulk propagator; the microscopic Hamiltonian generates boost time there.

5 A bilocal bridge to the first dynamical candidate

The first candidate can be written without numerical time derivatives. Differentiate the conserved M_1 using the product rule:

$$0 = \dot{M}_1 = \sum_i \dot{c}_i^\dagger \dot{c}_i + M_2, \qquad A_2 = -M_2 = \sum_i [H_J, c_i]^\dagger [H_J, c_i] \geq 0. \tag{10}$$

Thus A_2 is Hermitian in every sample. Center it with the same trace state, $O_2 = A_2 - d^{-1}\,\mathrm{Tr}(A_2) I$. Its nonzero-frequency spectral weight is our first concrete microscopic target. Positivity of A_2 is an operator identity; O_2 need not be positive after centering.

Rather than compute each derivative composite independently, organize the whole tower with a two-time operator,

$$B_J(\tau; t) = \sum_i c_i^\dagger(t) c_i(t+\tau), \qquad \partial_\tau^n B_J(\tau; t)|_{\tau=0} = M_n(t). \tag{11}$$

Define $b_J(\tau) = d^{-1}\,\mathrm{Tr}\, B_J(\tau; 0)$ and the connected-in-trace correlator

$$F_J(\tau, \sigma; t) = d^{-1}\,\mathrm{Tr}[B_J(\tau; t) B_J(\sigma; 0)] - b_J(\tau) b_J(\sigma). \tag{12}$$

At finite N the exact relation is

$$C_{22}(t) = \overline{\partial_\tau^2 \partial_\sigma^2 F_J(\tau, \sigma; t)|_{\tau=\sigma=0}}. \tag{13}$$

This turns the first composite two-point problem into a contracted fermion four-point problem with a specified operator order. The subtraction is made within each sample before averaging. Equation (13) is the interface to a real-time bilocal or ladder calculation, not yet a solution of that calculation. In the large-N limit its equal-time derivatives must be justified by control of the frequency moments.

How this guides the next calculation. We will derive the kernel for F_J in the chosen high-temperature limit, then take the derivatives in (13). A nonzero spectral component of O_2 is the first object to match to a bulk propagator. Centering removes the identity expectation; other conserved components are separated using the energy projectors in the appendix. For a quadratic number-conserving Hamiltonian, $A_2 = c^\dagger h^2 c$ is itself conserved. That limiting case provides a useful check that interactions, rather than the notation of the tower, generate the dynamics.

6 Research sequence and concrete deliverables

A. Complete the microscopic–geometric dictionary. Read [3, 4, 5, 8, 9] at equation level. Deliver one convention table: real versus complex N, interaction order, CAR, variance, H_c/H_s, proper time, entropy, and the two radial scales in (3). Reproduce the geometric redshift and the leading correlation times in a shared convention.

B. Derive the singlet correlation kernel. Work through the existing bilocal and ladder methods, starting with [11] and the real-time analysis in [7]. Construct the contracted channel before disorder averaging, keeping its J dependence. Deliver the kernel equation for C_{ab}, the conserved modes in (7), and the first nonzero moments for $n = 2, 3$, starting with the bilocal derivative in (13).

C. Calculate the spectrum in controlled limits. First validate the operator construction with finite-N diagonalization and the spectral sum rules. Start with small positive λ after double scaling, where [8], §5.3, proposes the meson comparison. Then organize corrections in λ: a fixed-order planar kernel alone does not sum the finite-λ expansion. Keep the Hamiltonian normalization fixed by the dictionary. Deliver the spectral matrix, its sample variation, and convergence with operator basis size.

D. Reconstruct particles and then interactions. Derive the bulk-to-horizon map in Rindler coordinates and compare the singlet correlators with the candidate gauge-theory propagators. The desired outputs are invariant mass ratios and spectral weights. Once that map is fixed, use three- and four-point functions to determine couplings and test the interaction interpretation of λ.

A Technical notes and present status

This is the first mainline construction and reading plan, not a completed singlet-spectrum computation. The normal-ordered identities and spectral sum rules above are standard CAR and linear-algebra consequences. The 2026 paper supplies the motivation, not the missing quantitative spectrum. Low-temperature conformal dimensions from [11] will be used as method references, not identified with infinite-temperature boost frequencies.

For finite samples the group action rotates both the fermions and the coupling tensor. The contracted family is covariant under this joint action; a fixed generic H_J has exact $U(1)$ charge conservation, not an assumed exact $U(N)$ symmetry. Conserved contributions to (8) sit at $\omega = 0$. They can be recorded separately using $\mathcal{P}_0 O = \sum_E P_E O P_E$, with P_E the full degenerate energy projector. This subtraction changes the measured probe, not the state or the Hamiltonian.

At exactly infinite temperature the trace of a commutator of two even operators vanishes. Equation (8) therefore starts with trace correlation spectra, not an ordinary bosonic retarded response. The bulk response and the observer map belong to step D. The earlier spectral-pentagon note is retained as a separate technical branch; it is not a prerequisite for this programme.

References

[1] L. Dyson, J. Lindesay and L. Susskind, *Is There Really a de Sitter/CFT Duality*, hep-th/0202163v4.

[2] H. Lin and L. Susskind, *Infinite Temperature's Not So Hot*, 2206.01083.

[3] L. Susskind, *De Sitter Space, Double-Scaled SYK, and the Separation of Scales in the Semiclassical Limit*, 2209.09999v1.

[4] A. A. Rahman, *dS JT Gravity and Double-Scaled SYK*, 2209.09997v2.

[5] L. Susskind, *De Sitter Space has no Chords. Almost Everything is Confined*, 2303.00792v1.

[6] A. Rahman and L. Susskind, *Infinite Temperature is Not So Infinite: The Many Temperatures of de Sitter Space*, 2401.08555v3.

[7] Y. Sekino and L. Susskind, *Double-Scaled SYK, QCD, and the Flat Space Limit of de Sitter Space*, 2501.09423v2.

[8] S. Miyashita, Y. Sekino and L. Susskind, *DSSYK at Infinite Temperature: The Flat-Space Limit and the 't Hooft Model*, 2506.18054v3.

[9] L. Susskind, *Where is the entropy in DSSYK–de Sitter? Correction to a Wrong Claim*, 2511.10907v1.

[10] S. Miyashita, Y. Sekino and L. Susskind, *Holograms and Standard Models*, 2607.05678v2.

[11] D. J. Gross and V. Rosenhaus, *The Bulk Dual of SYK: Cubic Couplings*, 1702.08016v2.